

## INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

### THE SITE

Location            Microsoft SAT 80, TX -- station KHD0, America/Chicago  
Committed pair    OSM 1533392813 -> 1533392812  
                     Microsoft SAT 81 / Microsoft SAT 80  
Facade gap        74.4 m between the two halls  
Weather record    43,148 real hourly records from KHD0  
Plume physics     576 steady-state solves on this site's own footprints; worst intake rise  
                     0.4822 C at 125 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

### THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day                    2022-02-13    (crossing)  
Plant limit            18.0 C  
Notice required        3 h  
Forecast skill          0.50 relative to persistence  
Level anchor           sensor  
Condenser bank        longest facade  
Switch budget          1 mode changes per day  
Minimum dwell          3 h  
Max dew point          15.0 C (Green Grid WP#46 p.6)

### WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 14 hours -- 11 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 15 hours were safe but still ran chillers, because the schedule could not afford them.

Agent                3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)  
Incumbent            14 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)  
                     the incumbent broke its own switch budget 1 time(s) to stay safe  
                     15 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

### EVERY HOUR, AND THE REASON FOR IT

| hh | mode       | safe | binding       | bound  | limit | actual | margin |
|----|------------|------|---------------|--------|-------|--------|--------|
| 00 | mechanical | yes  | switch budget | -1.819 | 18.0  | -2.220 | 1.600  |
| 01 | mechanical | yes  | switch budget | -1.290 | 18.0  | -1.110 | 1.434  |
| 02 | mechanical | yes  | switch budget | -1.675 | 18.0  | -1.718 | 1.214  |
| 03 | mechanical | yes  | switch budget | -2.077 | 18.0  | -3.280 | 1.444  |
| 04 | mechanical | yes  | switch budget | -1.192 | 18.0  | -2.984 | 1.314  |

|    |            |     |               |        |      |        |       |
|----|------------|-----|---------------|--------|------|--------|-------|
| 05 | mechanical | yes | switch budget | -1.497 | 18.0 | -2.984 | 1.179 |
| 06 | mechanical | yes | switch budget | -3.838 | 18.0 | -5.610 | 0.969 |
| 07 | mechanical | yes | switch budget | -1.722 | 18.0 | -2.984 | 1.233 |
| 08 | mechanical | yes | switch budget | 4.189  | 18.0 | 4.441  | 2.269 |
| 09 | mechanical | yes | switch budget | 7.763  | 18.0 | 9.450  | 3.113 |
| 10 | mechanical | yes | switch budget | 10.873 | 18.0 | 12.128 | 3.331 |
| 11 | mechanical | yes | switch budget | 14.139 | 18.0 | 13.804 | 2.209 |
| 12 | mechanical | yes | switch budget | 16.553 | 18.0 | 15.296 | 1.712 |
| 13 | mechanical | yes | switch budget | 17.675 | 18.0 | 16.584 | 1.429 |
| 14 | mechanical | no  | dry-bulb      | 18.776 | 18.0 | 17.691 | 1.518 |
| 15 | mechanical | no  | dry-bulb      | 19.135 | 18.0 | 17.694 | 1.476 |
| 16 | mechanical | no  | dry-bulb      | 25.404 | 18.0 | 31.312 | 1.234 |
| 17 | mechanical | no  | dry-bulb      | 25.020 | 18.0 | 29.914 | 1.519 |
| 18 | mechanical | yes | -             | 15.318 | 18.0 | 11.509 | 1.844 |
| 19 | mechanical | no  | dry-bulb      | 19.960 | 18.0 | 8.729  | 1.801 |
| 20 | mechanical | no  | dry-bulb      | 18.127 | 18.0 | 8.181  | 1.831 |
| 21 | FREE       | yes | -             | 7.943  | 18.0 | 4.736  | 2.068 |
| 22 | FREE       | yes | -             | 5.718  | 18.0 | 2.516  | 1.960 |
| 23 | FREE       | yes | -             | 3.911  | 18.0 | 0.000  | 1.447 |

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin  
actual = what the intake temperature turned out to be, from the station record  
margin = group-conditional forecast error for that hour of day, plus plume spread

## THE REASONING, HOUR BY HOUR

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### 00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.819 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.290 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.675 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.077 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.192 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a

thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**05:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.497 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**06:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.838 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**07:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.722 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**08:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.189 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**09:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.763 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**10:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.873 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**11:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.139 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**12:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.553 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is

also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**13:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.675 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**14:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 18.776 C against a 18.0 C limit, so it fails by 0.776 C. It would take a limit 0.776 C higher, or a bound 0.776 C tighter, to change this hour.

**15:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 19.135 C against a 18.0 C limit, so it fails by 1.135 C. It would take a limit 1.135 C higher, or a bound 1.135 C tighter, to change this hour.

**16:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 25.404 C against a 18.0 C limit, so it fails by 7.404 C. It would take a limit 7.404 C higher, or a bound 7.404 C tighter, to change this hour.

**17:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 25.020 C against a 18.0 C limit, so it fails by 7.020 C. It would take a limit 7.020 C higher, or a bound 7.020 C tighter, to change this hour.

**18:00 MECHANICAL**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.318 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**19:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 19.960 C against a 18.0 C limit, so it fails by 1.960 C. It would take a limit 1.960 C higher, or a bound 1.960 C tighter, to change this hour.

**20:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 18.127 C against a 18.0 C limit, so it fails by 0.127 C. It would take a limit 0.127 C higher, or a bound 0.127 C tighter, to change this hour.

**21:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 7.943 C, which is 10.057 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.068 C, and the margin is measured, not chosen: 1.641 C of group-conditional forecast error for this hour of day, plus 0.4266 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**22:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 5.718 C, which is 12.282 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.960 C, and the margin is measured, not chosen: 1.534 C of group-conditional forecast error for this hour of day, plus 0.4266 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**23:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 3.911 C, which is 14.089 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.447 C, and the margin is measured, not chosen: 1.151 C of group-conditional forecast error for

this hour of day, plus 0.2955 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.  
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